



Comparative efficacy of allopathic and herbal drugs in sheep naturally infected with coccidiosis

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ABSTRACT

Development of anticoccidial resistance and concerns of drug residues have prompted the evaluation of alternatives to allopathic drugs. In current study, anticoccidial effect of amprolium was compared with that of *Curcuma longa* and *Zingiber officinale*. Ninety (90) sheep, naturally infected with *Eimeria* spp. and having a minimum oocyst per gram (OPG) count of faeces above 5000 were randomly selected and divided into six groups of 15 animals each. Animals were supplemented with amprolium @ 62.50 mg/kg body weight (bw) (GI), turmeric @ 200 and 300 mg/kg bw (GII and GIII) and ginger @ 200 and 300 mg/kg bw (GIV and GV), orally for 7 days and GVI animals were kept as untreated infected control. Faecal samples were collected on '0' day before treatment and on 8th, 14th, 21st and 28th day after starting treatment and evaluated using Faecal oocyst count reduction test (FOCRT). The efficacy of amprolium was 93.18%, 96.82%, 95.56% and 95.80% on 8th, 14th, 21st and 28th day, after starting treatment. Turmeric @200 mg/kg b.w. showed efficacy of 41.49%, 52.37%, 61.47% and 60.08% and turmeric @ 300 mg/kg bw was 44.92%, 54.32%, 64.21% and 61.95% effective on 8th, 14th, 21st and 28th day, respectively. Ginger @200 mg/kg bw showed efficacy of 38.51%, 53.48%, 55.38% and 55.53% and ginger @ 300 mg/kg bw was 39.65%, 54.81%, 57.18% and 58.22% effective on 8th, 14th, 21st and 28th day, respectively. The results justify use of amprolium for clinical coccidiosis while *Curcuma longa* and *Zingiber officinale* could be used as natural prophylactic alternatives.

1. Introduction

The parasitic infections, especially gastrointestinal parasitism, is present more often as subclinical infection causing substantial economic losses without being noticed clearly. Among the parasitic diseases, coccidiosis is one of the critical diseases affecting the animal industry and is clinically associated with reduced growth, dehydration, fever, inappetence, weight loss, dysentery, anaemia, death in young animals and even wool breaking especially in sheep (Yakhchali and Golami, 2008; El-Sayed et al., 2020). The majority of the coccidia of importance in domestic animals belong to the genus *Eimeria* of Phylum Apicomplexa that cause tremendous damage to the intestinal mucous membrane. Small ruminants of all age groups and breeds are susceptible to *Eimeria* infection. Lambs aging from 3 weeks to 5 months are most severely affected, while the rest of the flock might act as carriers (Tauseef-ur-Rehman et al., 2011). Ovine coccidiosis is one of the most common causes of enteritis in sheep. Due to ubiquitous presence of eimerian

oocysts in and around animal's environment, it is impractical to avoid infections, therefore, oocysts of *Eimeria* spp. are generally present in small numbers in the faeces of healthy sheep of all categories of ages. (Skirnisson, 2007. Nadia et al., 2021). Disease outbreaks occur in susceptible animals when exposed to infection with pathogenic species. Prevention of losses and reduced productivity from coccidiosis depends on effective treatment to reduce environmental contamination by infective oocysts. Currently, chemotherapy forms the basis and is used extensively to control coccidiosis. However, the worldwide development of drug resistance and escalating costs of developing new drugs (Hur et al., 2005) have significantly reduced the commercial incentive to develop new chemical anticoccidials (Peek and Landman, 2003; Aljedaie and Al-Malki, 2020). Further, these anticoccidials are effective only against certain stages of the parasite, require long treatment period and have residual effects. Using phyto-genic feed additives as alternatives may be the best solution for the rural farmers to these growing concerns.

Resurgence of usage of medicinal plants to treat parasitism in

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ruminants has been seen of late to varying degrees of success. The herbal plants being eco-friendly with low cost of production have attracted the attention of various research workers and are becoming priority in most parts of the world (Castagna et al., 2021). The use of seeds, leaves, stem and dried rhizomes of many plants such as turmeric, mint, garlic, ginger have been explored to treat animals against gastrointestinal parasitism in the recent decade (Adeniji et al., 2017). Anticoccidial nature of certain herbal products has been reported by Hur et al. (2005), Abbas et al. (2010), Maria et al. (2016) and Adeniji et al. (2017) and they have shown promising results. Although, the essential biological properties of turmeric and ginger rhizomes make them potential anti-coccidial feed additives/, their use in this part of the globe has not been adequately documented. Only two studies on the use of anticoccidials and their comparison with phytomedicine had been carried out in goats, while no such study has been done in sheep of Jammu and Kashmir.

2. Material and methods

2.1. Study area and design

The study was conducted at the Mountain Research Centre for Sheep and Goat (MRCSG), SKUAST-K, Shuhama, Srinagar located in central Kashmir. The general topography of the area is both mountainous and plain. The climate is temperate with the upper-reaches receiving heavy snowfall in winter, with an average annual rainfall of 175 mm. No coccidiostats had been used in any of the study animals before the start of the trial. Faecal samples were collected a day before the beginning of the trial (Day 0) and examined to quantify the number of *Eimeria* oocysts per gram (OPG) of faeces. The trial included 90 animals naturally infected with *Eimeria* spp. with OPG values above 5000 which is considered significant for clinical coccidiosis in ruminants (Mohamed et al., 2009) which were randomly divided into six groups (G-I to G-VI) of 15 animals each: sheep that received amprolium @ 62.50 mg/kg body weight (G-I), sheep supplemented with turmeric @ 200 mg/kg body weight (G-II), sheep given turmeric @ 300 mg/kg body weight (G-III), sheep that received supplementation of ginger @ 200 mg/kg body weight (G-IV) and sheep given ginger @ 300 mg/kg body weight (G-V), orally for 7 days in each group. One group remained the untreated infected control (positive control) (G-VI). The dried rhizomes of both plants i.e., turmeric and ginger were purchased from local commercial sources and cut into pieces before being pulverized with an electric grinder to ensure purity.

2.2. Determination of therapeutic efficacy

To determine the therapeutic efficacy of different drugs, faecal samples from rectum were directly collected on "0" day before treatment and kept in refrigeration until their analysis and on 8th, 14th, 21st and 28th day after starting treatment. Samples were examined using the faecal floatation technique in order to confirm the presence of *Eimeria* spp. oocysts and parasitic load quantification i.e., OPG (oocyst per gram) was ascertained by modified McMaster's technique. Further, *Eimeria* oocysts were speciated by morphology and sporulation time of oocysts. The faecal oocyst count reduction test (in vivo test) was finally employed to calculate the percent efficacy by using arithmetic mean of the group OPGs as described by Thrushfield, 2005.

Efficacy of drug treatment % =

$$\frac{\text{Mean pre-treatment OPG} - \text{Mean post-treatment OPG}}{\text{Mean pre-treatment OPG}} \times 100$$

2.3. Statistical analysis

The results were subjected to standard statistical analysis as per Snedecor and Cochran (1994). Student's 't' test was employed for analysis of two means of OPG and more than two means of OPG were

compared through ANOVA.

3. Results

Amprolium is a commonly used anti-coccidial drug in small ruminants while as turmeric and ginger are routinely used in human food as well as in ethno-veterinary medicine. Comparison of anti-coccidial efficacy of amprolium vis-à-vis herbal alternatives turmeric and ginger was carried out in this study. All the animals were naturally infected with one or more species of *Eimeria*. *E. ahsata*, *E. granulosa*, *E. ovinoidalis*, *E. parva*, and *E. intricata* were identified in the pooled culture of faecal samples. (Fig. 1).

3.1. Efficacy of amprolium (Tables 1 and 2)

Group I animals treated with amprolium at the dose rate of 62.50 mg/kg body weight orally had a pre-treatment mean OPG count of faeces on Day 0 as 7410.0 ± 660.6 (5650–14,500). The mean OPG decreased to 486.67 ± 80.5 (250–1350), 226.67 ± 47.0 (0–650), 316.67 ± 57.6 (0–700) and 300.0 ± 66.3 (0–850), 0.00 $\pm$ 0.00 on 8th, 14th, 21st and 28th day, respectively, after starting treatment. It was found that the efficacy of the drug against *Eimeria* species was 93.18%, 96.82%, 95.56% and 95.80% on 8th, 14th, 21st and 28th day, respectively. On 8th day, normal faecal consistency was observed in animals that had semi-solid to loose consistency of faeces before starting the treatment.

3.2. Efficacy of turmeric @ 200 mg/kg b.w. (Tables 1 and 2)

Mean OPG count of faeces on 8th, 14th, 21st and 28th day was 4606.7 ± 491.6 (3000–9050), 3750.0 ± 391.6 (2400–7800), 3033.3 ± 350.4 (2100–6350) and 3143.3 ± 341.7 (1950–6000), respectively, while as pre-treatment mean OPG on day 0 was 7873.3 ± 633.6 (5700–13,500). Efficacy of turmeric @ 200 mg/kg b.w was found to be 41.49%, 52.37%, 61.47% and 60.08% on 8th, 14th, 21st and 28th day, respectively. Three animals of the group had semi-solid faeces before treatment among which two had normal consistency on the 14th day after starting treatment. However, the third animal showed no improvement in consistency up to the 21st day.

3.3. Efficacy of turmeric @ 300 mg/kg b.w (Tables 1 and 2)

Pre-treatment mean OPG count of faeces on day 0 was 7516.7 ± 677.4 (5550–15,150). Mean OPG count on the 8th, 14th, 21st, and 28th day was 4140.0 ± 429.9 (2900–8300), 3433.3 ± 376.5 (2300–7400), 2690.0 ± 291.9 (1900–5750) and 2860.0 ± 299.6 (1750–5900), respectively. It was found that the efficacy of the drug was 44.92% on the 8th day, 54.32% on day 14, 64.21% on day 21 and 61.95% on 28th day. Faecal consistency of five animals ranged from semi-solid to loose faeces, among which two showed normal consistency 14th day onwards.

3.4. Efficacy of ginger @ 200 mg/kg b.wt (Tables 1 and 2)

Pre-treatment mean OPG count of faeces on Day 0 was 6993.3 ± 567.3 (5600–12,050). Mean OPG count on the 8th, 14th, 21st and 28th day was 4300.0 ± 345.2 (2400–9500), 3253.3 ± 301.4 (2100–6050), 3120.0 ± 279.5 (2000–5850) and 3110.0 ± 215.0 (2150–5000), respectively. Therefore, efficacy of the drug was 38.51% on the 8th day, 53.48% on day 14, 55.38% on day 21st and 55.53% on the 28th day. All the animals had normal faecal consistency during the whole study period.

3.5. Efficacy of ginger @ 300 mg/kg b.w (Tables 1 and 2)

Pre-treatment mean OPG count of faeces on day 0 was 6750.0 ± 503.6 (5650–12,550). Mean OPG count on the 8th, 14th, 21st and 28th

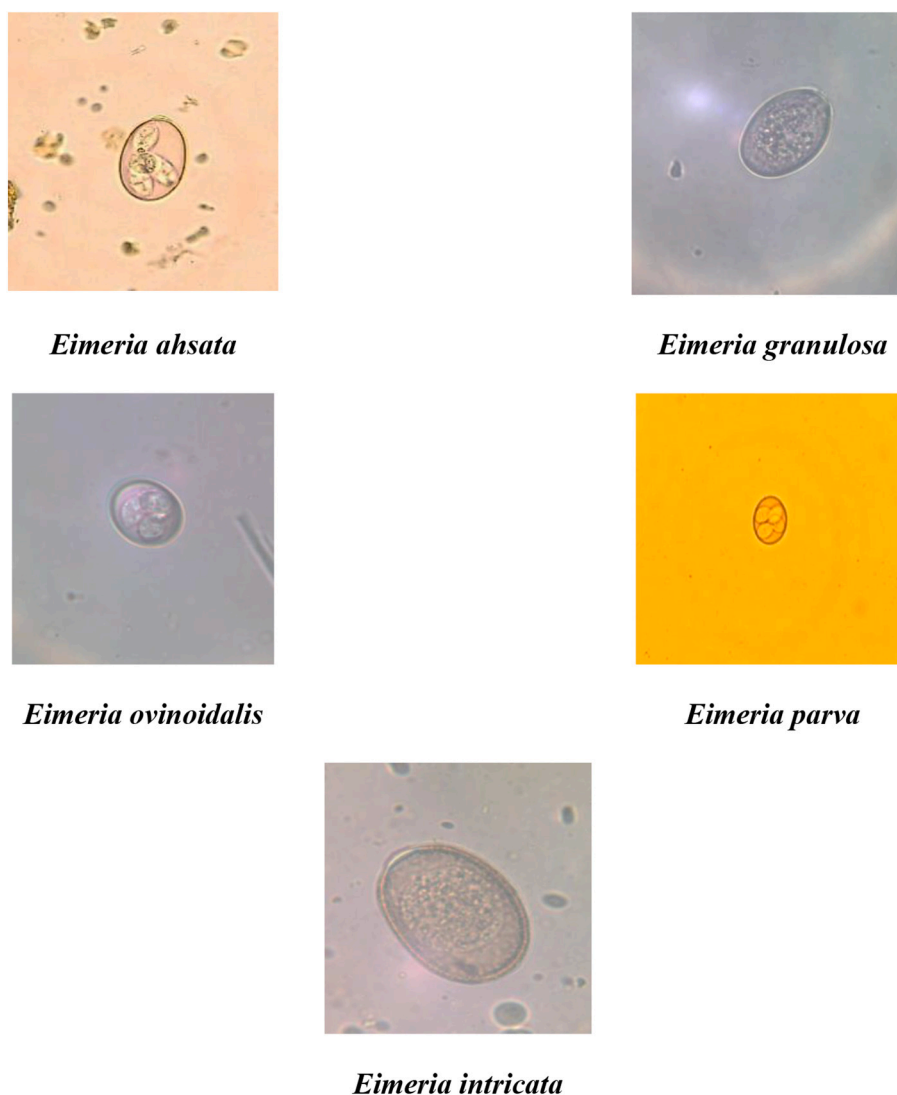


Fig. 1. Various *Eimeria* spp. oocysts obtained during the study (400X).

Table 1

Mean oocysts per gram (OPG) of faeces in sheep and percent reduction in OPG for different treatment groups observed on different days.

Group	Drug	Pre-treatment mean OPG	Post-treatment mean OPG at days			
			8	14	21	28
I (n = 15)	Amprolium (62.5 mg/kg bw)	7410.0 ± 660.6 ^a (5650–14,500)	486.67 ± 80.5 ^a (250–1350)	226.67 ± 47.0 ^a (0–650)	316.67 ± 57.6 ^a (0–700)	300.0 ± 66.3 ^a (0–850)
II (n = 15)	Turmeric (200 mg/kgbw)	7873.3 ± 633.6 ^a (5700–13,500)	4606.7 ± 491.6 ^b (3000–9050)	3750.0 ± 391.6 ^b (2400–7800)	3033.3 ± 350.4 ^b (2100–6350)	3143.3 ± 341.7 ^b (1950–6000)
III (n = 15)	Turmeric (300 mg/kgbw)	7516.7 ± 677.4 ^a (5550–15,150)	4140.0 ± 429.9 ^b (2900–8300)	3433.3 ± 376.5 ^b (2300–7400)	2690.0 ± 291.9 ^b (1900–5750)	2860.0 ± 299.6 ^b (1750–5900)
IV (n = 15)	Ginger (200 mg/kgbw)	6993.3 ± 567.3 ^a (5600–12,050)	4300.0 ± 345.2 ^b (2650–7000)	3253.3 ± 301.4 ^b (2100–6050)	3120.0 ± 279.5 ^b (2000–5850)	3110.0 ± 215.0 ^b (2150–5000)
V (n = 15)	Ginger (300 mg/kgbw)	6750.0 ± 503.6 ^a (5650–12,550)	4073.3 ± 509.2 ^b (2400–9500)	3050.0 ± 387.6 ^b (1900–6900)	2890.0 ± 359.5 ^b (1850–6350)	2820.0 ± 322.9 ^b (1450–6100)
VI (n = 15)	Control	7396.7 ± 419.0 ^a (6100–11,100)	13,000.0 ± 933.0 ^c (8000–21,500)	15,653.0 ± 973.8 ^c (9500–24,050)	15,473.0 ± 1263.5 ^c (8500–25,500)	12,760.0 ± 1208.5 ^c (7550–21,500)

Values in the same column bearing similar superscripts are statistically non-significantly different ($P > 0.05$) (Mean ± S.E).

day was 4073.3 ± 509.2 (2400–9500), 3050.0 ± 387.6 (1900–6900), 2890.0 ± 359.5 (1850–6350) and 2820.0 ± 322.9 (1450–6100), respectively. It was found that the drug's efficacy was 39.65% on the 8th day, 54.81% on day 14, 57.18% on day 21st and 58.22% on the 28th day. Of the three animals that showed semi-solid faecal consistency among which one had the presence of mucous also, two showed

improvements on the 14th day after starting treatment.

3.6. Control group (Table 1)

In the control group (positive control), the mean OPG of faeces for *Eimeria* species was found to be 7396.7 ± 419.0 day before starting the

Table 2

Efficacy of amprolium, turmeric and ginger against *Eimeria* species of sheep in Central Kashmir.

Group	Drug/Dose	Efficacy (%) of drugs on days			
		Day 8	Day 14	Day 21	Day 28
I	Amprolium @ 62.5 mg/kgbw	93.18 ^a	96.82 ^a	95.56 ^a	95.80 ^a
II	Turmeric @ 200 mg/kgbw	41.49 ^a	52.37 ^{ab}	61.47 ^b	60.08 ^b
III	Turmeric @ 300 mg/kgbw	44.92 ^a	54.32 ^{ab}	64.21 ^b	61.95 ^b
IV	Ginger @ 200 mg/kgbw	38.51 ^a	53.48 ^b	55.38 ^b	55.53 ^b
V	Ginger @ 300 mg/kgbw	39.65 ^a	54.81 ^{ab}	57.18 ^b	58.22 ^b

Values in the same row bearing similar subscript letters are statistically non-significantly different ($P > 0.05$).

treatment, which increased to $13,000.0 \pm 933.0$, $15,653.0 \pm 973.8$, $15,473.0 \pm 1263.5$ and $12,760.0 \pm 1208.5$ on 8th, 14th, 21st and 28th day, respectively.

On Day 0, the difference was not significant ($P > 0.05$) in mean OPG values of sheep among different groups. On the other hand, on day 8th, 14th, 21st and 28th ($P < 0.05$), in groups I, II, III, IV and V, mean OPG count decreased, while it increased in group VI significantly ($P < 0.05$). The difference was statistically significant between group I with all other groups ($P < 0.05$), while as groups II, III, IV and V showed non-significant ($P > 0.05$) difference within them.

No significant difference was observed in efficacy of amprolium at different time points ($P > 0.05$), while as turmeric at both the doses showed significant difference of efficacy on day 8 with day 21st and 28 ($P < 0.05$). Ginger showed a significant difference on efficacy on day 8 with all other days at 200 mg/kg body weight ($P < 0.05$) and with day 21 and 28 @ 300 mg/kg b.w. ($P < 0.05$). Therefore, the maximum efficacy of turmeric between the two doses was on 21st day after starting treatment @ 300 mg/kg body weight as 64.21%. Similarly, maximum effect of ginger was @ 300 mg /kg body weight as 58.22% on day 28.

4. Discussion

Our results regarding the efficacy of amprolium in sheep are in agreement with those reported by Abakar et al. (2005), who reported that amprolium @ 5 g/ 20 kg b.w reduced the faecal oocyst count by 98.44%, 95.02% and 95.85% on day 14, 21 and 28, respectively. Similarly, according to Soulsby (1982) amprolium @ 50–62.5 mg/kg body weight for sheep for four days or longer results in a rapid reduction in oocyst output and clinical recovery. An outbreak in a stall-fed sheep farm at Shimoga, treated with amprolium (55 mg/kg b. wt. daily PO for 7 days) led to recovery from the infection and faecal samples were found negative for oocysts of *Eimeria* spp. (Adeppa et al., 2016). The anticoccidial efficacy shown by amprolium in our study also provided an idea about lack of development of resistance to it as of now. In the present study, amprolium was found to be highly effective anticoccidial chemotherapeutic agent in terms of early reduction of oocysts and clinical recovery compared to herbal plants, thus reducing the parasitic pressure and the risk of further heavy contamination within the flock. This is especially important in a confined environment where many oocysts can accumulate in the bedding.

A novel idea for management and treatment of coccidiosis is the use of phyto-genic feed additives as an alternative to infeed medicines due their encompassing beneficial properties that include being of natural origin, minimal or no side effects, no residual effects in animal tissue and cost-effectiveness (Alzahrani et al., 2016; Alhotan and Abudabos, 2019; Al-Quraishy et al., 2020; Qasem et al., 2020). One potentially well-accepted alternative to chemical additives is turmeric (*Curcuma longa*), which is recognised to possess a strong potential as anticoccidial preparation (Khalafalla et al., 2011; Aljedaie and Al-Malki, 2020). The anticoccidial effect of turmeric crude powder may be attributed to its curcumin content, a natural polyphenolic compound abundant in the rhizome of *C. longa*. Turmeric helps in reducing shedding of oocysts and

lesion scores and has been reported to have ameliorative effects when given as additive to combat broiler coccidiosis caused by *Eimeria* spp. (Kim et al., 2013). Results obtained in the present study also show that *C. longa* significantly reduced oocyst production in lambs. As far as we know, this is the first report of *C. longa* used to treat *Eimeria* spp. infection in sheep in India. The action mechanism was not determined; however, it is reasonable to speculate that it produces effects through a direct toxic effect on coccidia, inducing apoptosis by the presence of precipitates on sporozoite surface affecting its morphology and viability and adhesion ability (Chattopadhyay et al., 2004). Similarly, ginger contains several compounds such as gingerol, gingerdiol, shogaols and gingerdione that possess anti-oxidant and anti-coccidial properties. Fresh ginger root and alcoholic extracts of ginger have established antioxidative capacities in vitro and live animals (Anonymous, 2012). Phenols cause impairment of crucial processes in the coccidia cells and finally their death by interacting with cytoplasmic membranes and changing their cation permeability. (Nasution et al., 2018; Aljedaie and Al-Malki, 2020).

In addition to the antiprotozoal effects of ginger and turmeric, antioxidant properties have also been demonstrated because oxidative stress is a hallmark of intracellular infections like coccidiosis. Considering alterations in the quantity of physiological antioxidants such as vitamins A, C and E in sheep suffering from coccidiosis; intestinal epithelial inflammation and increased tissue and blood concentrations of reactive oxygen species, such anti-oxidant effects of turmeric and ginger attain relevance (Maria et al., 2016). Overall, all these effects help counteracting damage caused by *Eimeria* spp. to integrity of epithelial cells and cellular oxidative injury of intestinal mucosa.

Various studies on the anticoccidial efficacy of these two drugs have been conducted in poultry, while only a few studies on sheep have been carried out worldwide. Maria et al., 2016 reported 100% efficacy in lambs treated orally for 14 days with 200 mg/kg body weight of *C. longa* on day 42 in Mexico. Similarly, the efficacy of the aqueous extract of curcumin (*C. longa*) @ 40 mg/kg body weight on the excretion of oocysts of *Eimeria* spp. in 24 New Zealand white rabbits was 80.1, 63.7 and 64.9% for day 28, 35 and 42, respectively (Maria et al., 2016). El-Khtam et al. (2014) evaluated the anticoccidial effect of turmeric powder in broilers and revealed anticoccidial activity comparable with that of the anticoccidial drug evidenced by reducing mortality, lowered gut lesion scores and reduced faecal oocyst shedding of the infected chickens as compared to the infected non treated group. Adeniji et al. (2017) reported the efficacy of crude powder of ginger rhizome as 100, 100, 98.83 and 100% on the 28th day @ 5 g/kg, 10 g/kg, 15 g/kg and 20 g/kg to Yankasa rams of Nigeria. Ali et al. (2019) reported the efficacy of ginger to be more in comparison to amprolium on coccidiosis in broiler chickens in Pakistan. The discrepancy among these studies is likely due to the different animals, physiological stages, diet compositions, drug source and its application level.

5. Conclusions

This is the first study demonstrating and comparing the efficacy of amprolium and beneficial actions of drenching *Curcuma longa* and *Zingiber officinale* to control coccidiosis of sheep in this part of the globe. Significantly lower OPG count was observed in amprolium treated group, moderate OPG count with progressive decline in *Curcuma longa* and *Zingiber officinale* treated groups and highest in the control group, indicating the substantial effect of amprolium and herbal plants in reducing the oocyst output. At the same time, although herbals treated animals started showing clinical improvement (if present), oocysts continued to be in faeces upto the 28th day of observation, indicating that the herbs were less efficacious compared to amprolium. Therefore, it may be concluded that amprolium could still be used in coccidiosis control programmes in Central Kashmir due to its high efficacy, more so in clinical coccidiosis while as ginger and turmeric provide promising alternatives to amprolium especially to be used as natural prophylactics.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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